Ashwin Chembu

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PROFILE

Applied-AI engineer building retrieval-augmented products with document ingestion, vector search, evidence extraction, and cited answers. Connects grounded AI workflows to usable interfaces and backend services.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

AI / Retrieval: RAG, LangChain, OpenAI, Azure OpenAI, Pinecone, FAISS

Mechanisms: Document ingestion, vector embeddings, intent routing, evidence extraction, citations

Product / Backend: React, FastAPI, Node.js, Express, MongoDB, Redis, AWS S3

Validation: pytest, grounded answers, generator/critic refinement, human review

EXPERIENCE

Business Technology Solutions Associate | ZS Associates

South San Francisco, CA | Jun 2024-Present

- Built React and **FastAPI RAG** assistants with AWS document ingestion and Redis services to support enterprise knowledge-search workflows.
- Contributed to enterprise **RAG** evaluation and model workflows using **FastAPI,** Uvicorn, and pytest to validate applied-AI service behavior.
- Built Python and Flask endpoints connecting AWS S3 and Smartsheet APIs to automate document management and cloud data handoffs.
- Translated ambiguous stakeholder requirements into testable rules, API integrations, source-to-target mappings, and delivery documentation.

Founder and Lead Developer | Tallyrus

Jun 2023-Present

- Developed essay-feedback workflows with Python, **LangChain, Pinecone,** and OpenAI to support retrieval-augmented grading for educators.
- Built Flask services with MongoDB persistence and S3 document storage to manage application data and support essay-feedback workflows.
- Designed React interfaces, rubric workflows, and iterative feedback loops that let educators review essays with consistent scoring logic.

Data Analyst / Web Development Lead | Tech4Good at UCSC

Mar 2023-Present

- Contributed 158 attributed commits to a collaborator-owned FigJam repository, implementing TypeScript chat, polling, and export workflows.
- Implemented 20+ attributed Node/Express REST routes with MongoDB persistence and logging for shared chat, text entry, and polling.
- Built Excel/XLSX export workflows and Python/Pandas analyses over 43.3K anonymized interaction rows spanning 2.8K verified session IDs.

PROJECTS

Lume | AI Meeting Intelligence

Dec 2025-Present

- Designed a **RAG** engine with vector embeddings, intent routing, evidence extraction, and generator/critic refinement for citation-backed answers.
- Implemented reusable Project Brain knowledge bases that retain terminology and relationships from transcripts and seed files for grounded Q&A.
- Built React and Node/Express workflows with MongoDB and S3 to capture, store, search, and query meeting transcripts and project files.
- Built responsive React interfaces and WebSocket interactions connecting meeting capture, document search, and reusable project context.

Azure RAG | Document Search Proof of Concept

Public proof of concept

- Built document search with Python, Azure OpenAI embeddings, **LangChain,** and **FAISS** similarity search to retrieve relevant source passages.
- Implemented PDF/TXT/Markdown ingestion, text chunking, vector indexing, and source citation to ground interactive Q&A in document context.
- Documented Azure embedding and chat-model setup with a 30+ document corpus for repeatable retrieval and source-grounded Q&A workflows.

Private Filedrop | Secure File Exchange

2026

- Built password-protected uploads and API-key-protected Express interfaces with OpenAPI documentation for secure file-exchange workflows.
- Implemented local and S3-compatible storage, migration tooling, version history, and unguessable download links for durable file handoffs.
- Exposed MCP tools for file organization and retrieval, with HMAC-SHA256 webhooks and timing-safe credential checks for signed service events.

EDUCATION

University of California, Davis | **B.S. Computer Science** | **GPA: 3.7** Graduated Jun 2025

CERTIFICATIONS

AWS Certified Cloud Practitioner | Google Data Analytics Certificate

LINKS

Portfolio: https://ashchembu.com | LinkedIn: linkedin.com/in/ashwin-chembu-a5292b205 | GitHub: github.com/ashwinchembu